

CALLAGHAN VALLEY, BC, Canada

WMO: 716880

Lat: 50.144N

Lon: 123.111W

Elev: 2900

StdP: 13.22

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.2	10.5	-2.6	5.3	10.1	3.8	7.4	14.2	7.3	34.3	6.5	31.9	2.0	80	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.5	83.7	59.4	79.7	58.3	75.7	57.2	61.6	77.7	60.0	75.4	58.4	72.3	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.2	74.8	64.0	54.5	70.3	62.0	53.1	66.7	60.5	28.8	78.5	27.6	75.2	26.5	72.7	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.2	6.1	5.4	DB	0.6	89.4	5.6	3.2	-3.4	91.7	-6.7	93.6	-9.9	95.5	-14.0	97.8
			WB	0.0	64.0	5.6	2.2	-4.0	65.6	-7.3	66.8	-10.5	68.1	-14.5	69.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	30.0	30.0	33.8	38.7	47.1	52.0	59.7	60.3	52.9	42.8	33.3	27.6
	DBStd	12.98	6.23	7.22	5.51	5.13	6.16	6.51	6.53	6.45	7.12	5.58	6.53	6.47
	HDD50	3686	621	559	502	343	134	47	3	2	44	235	501	695
	HDD65	8308	1086	979	967	790	556	395	192	177	368	687	951	1160
	CDD50	922	0	0	0	3	44	107	303	320	133	12	0	0
	CDD65	68	0	0	0	0	5	27	30	6	0	0	0	0
	CDH74	1182	0	0	0	0	16	132	472	449	113	0	0	0
	CDH80	306	0	0	0	0	2	35	132	115	22	0	0	0
(14) Wind	WSAvg	2.0	1.4	1.8	2.1	2.4	2.5	2.5	2.5	2.2	2.0	1.8	1.6	1.3
(15) Precipitation	PrecAvg	36.1	4.1	2.6	3.0	2.2	2.0	2.7	2.0	2.0	2.4	3.8	5.0	4.3
	PrecMax	46.3	8.5	5.8	5.2	4.0	3.9	5.4	4.7	5.4	5.0	7.0	10.4	6.3
	PrecMin	28.3	2.2	0.1	0.6	0.9	0.8	1.3	0.4	0.5	0.3	0.4	2.0	2.0
	PrecStd	4.6	1.4	1.3	1.1	0.8	0.8	0.9	1.0	1.1	1.3	1.6	2.1	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	50.3	57.0	66.0	75.9	85.0	89.2	87.4	83.1	67.4	50.1	43.2
		MCWB	40.0	38.8	43.2	48.7	53.8	58.7	61.5	60.8	58.4	52.1	47.2	41.9
	2%	DB	39.8	44.0	49.6	57.3	70.7	77.7	83.5	83.4	77.2	60.1	46.1	38.5
		MCWB	37.3	37.4	40.5	43.5	52.5	57.3	60.1	59.3	56.8	49.6	43.8	37.6
	5%	DB	37.3	40.5	44.5	51.9	65.5	71.0	79.4	79.2	70.9	54.2	43.6	36.1
		MCWB	36.2	37.2	38.3	41.0	50.3	55.0	58.6	58.6	55.2	48.0	42.1	35.4
	10%	DB	35.5	37.5	41.0	47.2	60.1	65.2	75.0	74.7	64.0	50.3	41.3	34.3
		MCWB	34.6	35.5	36.8	39.3	48.6	52.7	57.7	57.6	53.3	46.0	40.2	33.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	41.5	45.8	50.1	57.0	60.9	64.3	63.0	61.1	54.8	48.7	42.3
		MCDB	42.8	44.5	53.5	64.3	69.2	78.5	82.7	78.7	76.8	60.8	49.9	42.8
	2%	WB	37.8	39.5	41.9	45.7	53.7	58.3	62.1	61.3	58.6	51.9	44.5	37.8
		MCDB	39.1	42.6	47.3	55.0	67.6	74.1	78.3	77.1	70.6	58.3	45.3	38.4
	5%	WB	36.2	37.4	39.3	42.3	51.4	55.7	60.1	59.9	56.7	49.1	42.3	35.6
		MCDB	37.2	39.5	43.3	49.2	62.9	68.9	75.8	75.2	67.0	53.4	43.4	36.0
	10%	WB	34.7	35.4	37.2	40.1	49.1	53.3	58.2	58.4	54.4	46.8	40.3	33.9
		MCDB	35.4	37.3	40.3	45.5	58.8	63.4	72.4	72.2	63.0	49.9	41.2	34.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.7	13.3	15.7	20.7	18.6	21.1	20.5	17.2	13.0	8.8	8.9
		MCDBR	10.3	15.9	20.1	25.3	30.5	30.0	30.0	29.1	27.9	21.1	11.8	9.3
	5% WB	MCWBR	7.8	10.0	11.2	12.5	14.6	13.4	11.3	10.1	11.4	11.9	8.8	8.0
		MCWBR	9.6	12.8	17.0	21.5	26.5	26.8	26.7	25.4	23.7	18.6	10.2	8.3
(40) Clear-Sky Solar Irradiance	taub	0.269	0.271	0.291	0.329	0.343	0.348	0.340	0.336	0.312	0.308	0.276	0.261	
		taud	2.264	2.292	2.244	2.151	2.243	2.350	2.402	2.442	2.517	2.489	2.373	2.265
	Ebn at Noon	245	278	291	289	289	288	289	285	283	262	242	229	
		Edh at Noon	22	28	35	43	41	38	35	32	27	23	20	19
(44) All-Sky Solar Radiation	RadAvg	272	546	799	1282	1683	1679	1877	1633	1122	626	306	219	
	RadStd	48	83	127	160	162	208	187	125	99	101	61	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page